One-Step Generated Effective-Action Transfer

Explicit retained/fiber recursion, centered transport, and defect sensitivity

Research dossier—restricted theorem and frontier audit

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Abstract

An explicit four-dimensional cubical block supplies a retained space of dimension 51, a conditional $\mathfrak{su}(2)$ fiber of dimension 357, and a certified tangent fiber lower bound $\lambda_F = 2$. The next question is whether a local nonlinear certificate can survive the operation that actually generates an effective action: integration of the fiber. This paper proves a finite-dimensional one-step theorem. A five-component scaled derivative norm controls fiber coercivity, conditional-minimizer motion, centered conditional transport, and the Hessian of the exactly marginalized action. The generated good-action budget is

$$\eta_g = \varepsilon_Y + \frac{\varepsilon_X^2}{\lambda_F - \varepsilon_F}.$$

A gauge-invariant bad/good ratio with scaled C^2 budgets contributes at most $\zeta_2 + \zeta_1^2$. State normalization and conditional observable transport are exact, and residual orthogonal covariance is preserved. These are PROVED implications from declared finite-chart hypotheses. Explicit Yang–Mills initialization of those hypotheses, dependent large-field control, locality across blocks, repeated ultraviolet iteration, and infrared mass generation remain OPEN.

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1 Target and epistemic boundary

The preceding quotient/fiber construction isolates two kinds of information. The retained variable y carries the coarse curvature data that must survive blocking; the fiber variable z carries the fluctuation information integrated out at one step. A local Hessian estimate on z is useful only if it can be converted into a controlled statement about the generated action on y.

The target here is therefore neither a formal beta function nor an infrared claim. It is the exact finite-dimensional implication

$$\text{input derivative certificate} \longrightarrow \text{fiber marginalization} \longrightarrow \text{output derivative certificate}.$$

Balaban's constructive programme generates localized effective actions in a small-field approximation through one-step transformations [1]; the theorem below abstracts one analytic mechanism needed inside such a step. The heat-kernel action motivating the initialization was introduced by Menotti and Onofri [3].

The theorem is a restricted-model result at claim level C2. It does not show that the four-dimensional nonlinear $SU(2)$ measure satisfies its input norm, that the exceptional set is summable, or that the recursion closes through a number of scales diverging with the cutoff. The Clay target remains the construction of nontrivial pure Yang–Mills theory on $\mathbb{R}^4$ with a positive Hamiltonian gap [5].

2 Typed one-step package

Let $Y = \mathbb{R}^m$ be the retained space and $F = \mathbb{R}^n$ the conditional fiber. For the block imported from the previous stage, $(m, n) = (51, 357)$. Let $D \subset F$ be a fixed convex domain and $t > 0$ the declared heat-time normalization. Write

$$S(y, z) = \frac{1}{2t}\big(\langle y, L_Y y\rangle + \langle z, L_F z\rangle\big) + R(y, z), \tag{1}$$

where

$$L_F \succeq \lambda_F \mathrm{I}, \qquad L_Y \succeq \lambda_Y \mathrm{I}. \tag{2}$$

The coordinates are Gaussian conditional-mean coordinates: the tangent quadratic form has no retained–fiber mixed term. This is a choice of local splitting, not a global gauge fixing theorem.

Define the good partition function, effective action, and conditional law by

$$Z_{\mathrm{g}}(y) = \int_D e^{-S(y,z)}\, dz, \quad \widehat{S}_{\mathrm{g}}(y) = -\log Z_{\mathrm{g}}(y), \quad \nu_y(dz) = Z_{\mathrm{g}}(y)^{-1} e^{-S(y,z)}\, dz. \tag{3}$$

All differentiations below are justified either on a bounded D or by the uniform fiber convexity and the standard domination hypotheses stated with the theorem.

Definition 2.1 (Scaled local action norm)**.** The remainder R has budget

$$\mathfrak{N}_t(R) = (\varepsilon_F, \varepsilon_X, \varepsilon_Y, \tau_F, \tau_X)$$

on a declared region if

$$t\|R_{zz}\| \le \varepsilon_F, \qquad t\|R_{zy}\| \le \varepsilon_X, \qquad t\|R_{yy}\| \le \varepsilon_Y. \tag{4}$$

$$t\|D_z R_{zz}\| \le \tau_F, \qquad t\|D_z R_{zy}\| \le \tau_X. \tag{5}$$

All norms are Euclidean operator or multilinear operator norms. The five entries are dimensionless and are part of the RG state; they are not hidden in big-O notation.

3 Fiber margin and centered transport

Set

$$\mu := \lambda_F - \varepsilon_F. \tag{6}$$

Proposition 3.1 (Fiber coercivity and minimizer response, PROVED)**.** *Assume $\mu > 0$. Then*

$$S_{zz}(y,z) \succeq \frac{\mu}{t}\mathrm{I}.$$

Whenever the conditional minimizer $z_\star(y)$ is interior, it is unique and

$$\|Dz_\star(y)\| \le h, \qquad h := \frac{\varepsilon_X}{\mu}. \tag{7}$$

Proof. Equations (2) and (4) give the Hessian bound. Differentiate $S_z(y, z_\star(y)) = 0$ to obtain

$$Dz_\star = -S_{zz}^{-1} S_{zy}.$$

The tangent mixed Hessian vanishes, so $S_{zy} = R_{zy}$. Hence $\|S_{zz}^{-1}\| \le t/\mu$ and $\|S_{zy}\| \le \varepsilon_X/t$, proving (7). □

For $w = z - z_\star(y)$, define the centered potential

$$V_y(w) = S(y, z_\star(y) + w) - S(y, z_\star(y)).$$

Proposition 3.2 (Explicit centered cross budget, PROVED)**.** *Suppose (5) holds on every segment $z_\star(y) + sw$, $0 \le s \le 1$, with $\|w\| \le r$. Then*

$$t\|D_y \nabla_w V_y(w)\| \le \chi(r) := r\left(\tau_X + \tau_F \frac{\varepsilon_X}{\mu}\right). \tag{8}$$

Proof. Differentiating $S_z(y, z_\star(y)) = 0$ also shows that $D_y \nabla_w V_y(0) = 0$. At a general w,

$$D_y \nabla_w V_y(w) = S_{zy}(y, z_\star + w) + S_{zz}(y, z_\star + w) Dz_\star.$$

Subtract its value at $w = 0$. The constant tangent fiber Hessian cancels. The mean-value theorem, (5), and (7) give (8). □

Let $B_r \subset F$ be a common centered convex tube, and normalize $e^{-V_y(w)}\,dw$ on B_r.

Corollary 3.3 (Centered-fiber transport, PROVED)**.** *For retained points connected by a segment inside the declared chart,*

$$W_2(\widetilde{\nu}_y, \widetilde{\nu}_{y'}) \le K_F(r)\|y - y'\|, \qquad K_F(r) := \frac{\chi(r)}{\mu}. \tag{9}$$

Thus $K_F(r) < 1$ is an explicit local centered-fiber contraction certificate.

Proof. The centered potentials are μ/t strongly convex. The Brascamp–Lieb/logarithmic-Sobolev transport argument bounds response by the ratio of the cross-gradient constant $\chi(r)/t$ to the convexity constant μ/t; see also [4]. Equation (9) follows. □

4 Exact marginalization and the generated Hessian

Theorem 4.1 (One-step generated-action bound, PROVED)**.** *Assume R is C^2, $\mu > 0$, differentiation under (3) is valid, and the Brascamp–Lieb covariance inequality applies to ν_y. Then*

$$-\left(\varepsilon_Y + \frac{\varepsilon_X^2}{\mu}\right) \mathrm{I} \preceq t\nabla_y^2 \widehat{S}_{\mathrm{g}}(y) - L_Y \preceq \varepsilon_Y \mathrm{I}. \tag{10}$$

In particular,

$$\left\| t\nabla_y^2 \widehat{S}_{\mathrm{g}}(y) - L_Y \right\| \le \eta_{\mathrm{g}}, \qquad \eta_{\mathrm{g}} := \varepsilon_Y + \frac{\varepsilon_X^2}{\mu}. \tag{11}$$

Proof. Twice differentiating the logarithm of the partition function gives the exact identity

$$\nabla_y^2 \widehat{S}_{\mathrm{g}} = \mathbb{E}_{\nu_y}[S_{yy}] - \mathrm{Cov}_{\nu_y}(S_y, S_y). \tag{12}$$

The deterministic part $t^{-1}L_Y y$ has zero covariance. For every unit vector $u \in Y$, the Brascamp–Lieb covariance inequality [2] gives

$$\begin{aligned} \mathrm{Var}_{\nu_y}(u \cdot R_y) &\le \mathbb{E}_{\nu_y}\left[\langle R_{zy}u, S_{zz}^{-1} R_{zy}u\rangle\right] \\ &\le \frac{\varepsilon_X^2}{t\mu}. \end{aligned}$$

Moreover, $-\varepsilon_Y \mathrm{I}/t \preceq R_{yy} \preceq \varepsilon_Y \mathrm{I}/t$. Insert these estimates in (12). Covariance is positive semidefinite, yielding both sides of (10) and hence (11). □

The term ε_X^2/μ is the nonlinear analogue of a Schur-complement correction. In a Gaussian model, the generated Hessian is exactly

$$A_{\mathrm{eff}} = A - BC^{-1}B^*.$$

The accompanying implementation checks this identity and its covariance under orthogonal retained and fiber recodings.

5 The defect channel

Let the full one-block partition function split into good and bad parts:

$$Z_{\mathrm{full}}(y) = Z_{\mathrm{g}}(y) + Z_{\mathrm{b}}(y) = Z_{\mathrm{g}}(y)(1 + \delta(y)), \qquad \delta(y) := \frac{Z_{\mathrm{b}}(y)}{Z_{\mathrm{g}}(y)} \ge 0.$$

Consequently,

$$\widehat{S}_{\mathrm{full}}(y) = \widehat{S}_{\mathrm{g}}(y) - \log(1 + \delta(y)). \tag{13}$$

Definition 5.1 (Scaled defect norm)**.** On the retained chart, suppose

$$0 \le \delta \le \zeta_0, \qquad \sqrt{t}\|D\delta\| \le \zeta_1, \qquad t\|D^2\delta\| \le \zeta_2. \tag{14}$$

The activity ζ_0 controls mass comparison, whereas ζ_1, ζ_2 control the generated derivative budget.

Proposition 5.2 (Defect sensitivity, PROVED)**.** *Under* (14)*,*

$$t \left\| D^2 \log(1+\delta) \right\| \le \eta_{\mathrm{b}} := \zeta_2 + \zeta_1^2. \tag{15}$$

Therefore

$$\left\| t\nabla_y^2 \widehat{S}_{\mathrm{full}} - L_Y \right\| \le \eta_{\mathrm{full}} := \varepsilon_Y + \frac{\varepsilon_X^2}{\lambda_F - \varepsilon_F} + \zeta_2 + \zeta_1^2. \tag{16}$$

If $L_Y \succeq \lambda_Y \mathrm{I}$ *and* $\eta_{\mathrm{full}} < \lambda_Y$*, the generated base action is at least* $(\lambda_Y - \eta_{\mathrm{full}})/t$ *strongly convex on this finite chart.*

Proof. Differentiate $\log(1+\delta)$ twice:

$$D^2 \log(1+\delta) = \frac{D^2\delta}{1+\delta} - \frac{D\delta \otimes D\delta}{(1+\delta)^2}.$$

Since $\delta \ge 0$, (15) follows from the triangle inequality. Combine it with Theorem 4.1 and (13). □

The proposition identifies exactly what a large-field theorem must supply. It does not prove the bounds (14). In particular, ζ_0 must eventually arise from a dependent polymer estimate rather than from treating blocks as independent.

6 State, observables, and covariance

For an integrable observable $O(y, z)$, define its blocked representative

$$(\mathcal{T}O)(y) = \int_D O(y,z)\, \nu_y(dz). \tag{17}$$

Fubini's theorem gives the exact state-transport identity

$$\int O(y,z) e^{-S(y,z)}\, dy\, dz = \int (\mathcal{T}O)(y) e^{-\widehat{S}_{\mathrm{g}}(y)}\, dy. \tag{18}$$

Thus marginalization preserves positivity and normalization and transports observables by conditional expectation. It is lossy as a map of individual configurations but exact as a state/observable duality.

Let a compact residual group H act orthogonally by $(y, z) \mapsto (\rho_Y(h)y, \rho_F(h)z)$. If D is invariant and $S(\rho_Y(h)y, \rho_F(h)z) = S(y, z)$, then a change of variables proves

$$\widehat{S}_{\mathrm{g}}(\rho_Y(h)y) = \widehat{S}_{\mathrm{g}}(y).$$

Equivariant observables remain equivariant under $\mathcal{T}$. This is the covariance of the local residual representation. A global gauge-orbit atlas, including exceptional stabilizers, is a separate open problem.

7 RG certificate and parameter registry

Certificate field	Stage-06 result	Qualification
Configuration spaces	finite $Y \oplus F$	local quotient/fiber chart
Blocking kernel	exact conditional integration	fixed declared domain
Gauge covariance	preserved for invariant orthogonal residual action	global orbit atlas open
State transport	exact, Eq. (18)	good channel; defect added by Eq. (13)
Observable transport	exact conditional expectation	support/locality growth not bounded
Measure positivity	preserved	weaker than OS reflection positivity
Physical scale	t explicit; block factor must be declared externally	no energy-unit mass conclusion
Locality	absent from the finite norm	polymer interaction norm open
Volume dependence	one finite block	uniform-volume theorem open
Running coupling	not extracted	initialization and scheme open
Reconstruction	conditional law ν_y retained	individual fiber configuration forgotten
Error recursion	Eqs. (9), (11), (16)	one step only

Parameter	Meaning	Required condition
λ_F	tangent fiber lower bound	imported value 2
ε_F	nonlinear fiber-Hessian budget	$\varepsilon_F < \lambda_F$
ε_X	retained–fiber mixed budget	finite
ε_Y	direct retained-Hessian budget	finite
τ_F, τ_X	fiber variation of second derivatives	finite on declared tube
r	centered good-tube radius	$K_F(r) < 1$ for contraction
$\zeta_0, \zeta_1, \zeta_2$	defect mass and derivative budgets	must be proved from full measure
λ_Y	finite-chart base reference curvature	not a physical gap

8 Executable calibration and hostile cases

The implementation evaluates the formulas without asymptotic notation. With $\lambda_F = 2$, $\lambda_Y = 1$, and the small example

$$(\varepsilon_F, \varepsilon_X, \varepsilon_Y, \tau_F, \tau_X) = (0.25, 0.20, 0.10, 0.30, 0.20),$$

$r = 0.5$, and $(\zeta_0, \zeta_1, \zeta_2) = (0.01, 0.02, 0.03)$, it returns

$$\mu = 1.75, \quad K_F \approx 0.06694, \quad \eta_{\mathrm{g}} \approx 0.12286, \quad \eta_{\mathrm{full}} \approx 0.15326.$$

A deterministic $3+4$ dimensional Gaussian witness has actual generated remainder norm 0.0937808, below the certified bound 0.107226.

The test suite also enforces the boundary of the theorem:

- $\varepsilon_F \geq \lambda_F$ fails closed;
- a large tube may retain fiber coercivity while losing $K_F < 1$;
- a nonpositive fiber Hessian and a nonorthogonal claimed frame are rejected;

- $\lambda_Y = 0$ is never converted into a positive mass claim;
- the Gaussian Schur complement is invariant under orthogonal retained and fiber recodings.

These tests mechanize finite algebra only. They do not sample gauge fields or certify the analytic initialization.

9 Frontier audit and restrategy

ID	Required implication	Status
R1	One-step derivative, transport, and defect recursion from declared norms	PROVED
F1	Explicit cutoff-aware initialization of all five derivative budgets for the exact nonlinear $SU(2)$ heat-kernel block and generated interaction	OPEN
F2	Gauge-invariant dependent polymer estimate proving the defect budgets, including topology and exceptional orbits	OPEN
F3	Uniform spatial locality, observable-support control, and positivity preservation or recovery	OPEN
F4	Closure of the recursion for $k(a) \to \infty$ steps with summable errors	OPEN
F5	Nonperturbative crossover of the retained base into a massive basin	OPEN
F6	Nontrivial regulator-independent OS continuum theory and Hamiltonian spectral transfer	OPEN

The strongest advance is representational: the local fiber certificate is now connected to the exact operation that forgets the fiber and generates the next description. The theorem explains which information survives and the exact price of forgetting it. It also localizes the first unproved implication. The obstruction is no longer the algebra of marginalization; it is proving that the nonlinear gauge theory enters the declared norm, including its bad channel, with constants stable in cutoff and volume.

The next target should therefore be an initialization theorem, not another abstract recursion. It should compute explicit $SU(2)$ heat-kernel/BCH chart constants for the five norm components and pair them with a genuinely gauge-invariant defect activity. If the resulting values violate $\varepsilon_F < 2$ or $K_F < 1$, the failure will identify whether the retained variables, chart, or good/bad decomposition must change.

References

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